

Practice Classifying Agent Types in Real-World Use Cases

Objective

This activity builds your ability to diagnose and differentiate between reactive, deliberative, and hybrid AI agents in the real world. By analyzing system behaviors and suggesting architectural improvements, you'll sharpen your understanding of how agent design affects capability. Are you spotting surface behaviors—or underlying intelligence? This is your step toward thinking like an AI systems architect.

Real-World Use Case

Company Context: Global Logistics Corp. operates a fleet of 500 delivery trucks across three major regional hubs.

The Problem: Over the last quarter, operational costs rose by 14%. An internal audit revealed two primary pain points:

1. **Route Inefficiency:** Drivers are manually selecting delivery sequences, resulting in an average of 22% redundant mileage overlap per route.
2. **Maintenance Downtime:** Trucks are only serviced *after* a mechanical breakdown occurs, leading to an average of 4.5 days of un-utilized vehicle downtime per incident, severely stalling local supply chains.

Real-World Use Case

The Objective: The company is implementing an automated route optimization software and transitioning to a predictive maintenance schedule (servicing vehicles based on mileage thresholds before they break down).

Assignment Task: Reference the "Global Logistics Corp." scenario details above to answer all subsequent application and analysis questions.

Instructions for learners

You'll classify six real-world AI applications, some clear-cut, others intentionally ambiguous, into reactive, deliberative, or hybrid categories.

For each case, write 1–2 sentences justifying your classification based on agent characteristics (e.g., memory, planning ability, sensor input).

Then, for two of your examples, suggest one improvement that could move the agent from reactive to hybrid, or from hybrid to more deliberative behavior.

Conclusion

This exercise is designed to sharpen your diagnostic skills and help you recognize the nuances in real-world agent architectures.